Gwynedd Mercy University-School of Business & Education
(*Must be used for each lesson observed by University Supervisor*)
Lesson Plan Framework

Subject Area: Learning Support Math A **Grade Level: 9**

Student Name: Grace Dittman

<table>
<tr><th colspan="2">Domain I: Planning and Preparation</th></tr>
<tr><td colspan="2">PA Standard(s):

Pennsylvania Academic Standards for Mathematics Algebra/Numbers and Quantities: students apply numerical reasoning, operations, and properties of real numbers. This lesson specifically addresses comparing/ordering real numbers, absolute value, and evaluating numerical expressions using the order of operations.</td></tr>
<tr><td colspan="2">Essential Question(s):
Students will consider and reflect on throughout the lesson:
How can I use absolute value and the order of operations to accurately solve and compare mathematical expressions?

Students will consider and reflect on throughout the lesson: How do I know which number is greater when numbers are written in different forms? Why does the order of operations matter when solving an expression?</td></tr>
<tr><td colspan="2">Objectives/Performance Expectations
What will students know and be able to do as a result of this lesson?

By the end of the lesson, students will: (1) sequence a set of numbers from least to greatest, including fractions, decimals, whole numbers, and square roots, using a calculator as needed; (2) determine the absolute value of a number and explain that it represents distance from zero; and (3) use the order of operations (PEMDAS) to evaluate numerical expressions accurately, using a calculator to check computations.</td></tr>
<tr><td>Assessment Evidence/Level of Learning
Students will show what they know by evidence of/list names of formative or summative checks, quizzes, tests which indicate evidence of learning:
Students will show what they know by: completing the ordering-numbers warm-up with their table group; responding to teacher questions during guided notes and examples; solving practice examples on their calculators; completing an independent absolute-value maze worksheet; completing an independent</td><td>Differentiated Instruction:
Modifications in instruction and assessment for students with learning differences, specifically those with 504’s and IEP’s.
Extensions/Enrichment in instruction and assessment for students who are above grade level and/or gifted.
Modifications in instruction and assessment for students with learning differences, specifically those with 504’s and IEP’s. Provide fill-in-the-blank notes,</td></tr>
</table>

order-of-operations color-by-number worksheet; and, if time permits, participating in a Blooket review. Teacher will circulate, question students, check calculator work, and provide immediate feedback	explicit modeling, chunked directions, repeated directions/check-ins, calculator access, extended time as needed, teacher/peer support during the group warm-up, reduced problem sets or selected problems when indicated by an IEP, and opportunities to demonstrate understanding orally or through guided work. Allow students to use familiar calculator procedures and visual reminders for PEMDAS. Extensions/Enrichment: students who demonstrate mastery may explain their strategy to a peer, compare multiple forms of the same number, or complete additional challenge expressions.

Domain II: Classroom Environment:

Describe Classroom/Behavior Management Plan & Routines

Students will enter, collect/use the provided warm-up materials, and begin the ordering-numbers activity with their table group. Expectations will be stated before the activity: work cooperatively, handle number cards appropriately, use calculators responsibly, explain thinking, and remain engaged. During direct instruction, students will follow along with the fill-in-the-blank notes and participate when called on. During independent practice, students will work quietly, ask for help when needed, and remain on the assigned task. Teacher will circulate throughout the room, reinforce appropriate participation, redirect off-task behavior, and provide proximity and individual support as needed.

Domain III: Instruction

Activate Prior Knowledge/Motivation:

Preparing students for the lesson: How will you activate prior knowledge, build background, or review previous lessons as well as motivate students for the lesson?

Begin with a collaborative ordering-numbers challenge. Students will receive individual number cards containing numbers in different forms (including whole numbers, decimals, fractions, and square roots) and work with their table group to arrange them from smallest to largest. Students may use calculators to determine decimal approximations and compare values. After groups finish, the class will discuss the sequence and explain how they determined the order. This activates prior knowledge of number sense, comparing values, decimals, fractions, and calculator use while creating a meaningful transition into absolute value and numerical expressions.

Materials Needed:
Printed number cards cut into individual squares; calculators; fill-in-the-blank guided notes; pencils; absolute-value maze worksheet; order-of-operations color-by-number worksheet; Blooket (if time permits); teacher computer/projector as applicable.

Vocabulary (New or Review):

absolute value, distance from zero, positive, negative, opposite, expression, operation, order of operations, PEMDAS, parentheses, exponent, multiplication, division, addition, subtraction, least to greatest.

Sequence of the Lesson:
What learning activities/strategies will you use to engage the students in the learning? What will students do to use and apply new concepts or skills (independent practice if relevant)? How will you monitor and guide their performance? Include relevant vocabulary. Use bullets to sequence your lesson.

• 0–8 minutes — Entry/Ordering Numbers Warm-Up: Students enter and work with their table group to arrange the provided number cards from smallest to largest. Cards may include whole numbers, decimals, fractions, and square roots. Students use calculators as needed. Teacher circulates, observes strategies, asks questions, and supports students who need help comparing values.

• 8–15 minutes — Class Review of Warm-Up: Review the sequence as a class. Ask students how they compared fractions, decimals, square roots, and whole numbers. Address misconceptions and model calculator strategies when helpful. Connect the activity to the day's objective of accurately working with numbers and expressions.

• 15–25 minutes — Absolute Value Guided Notes: Distribute the fill-in-the-blank notes. Introduce absolute value as the distance of a number from zero on a number line. Complete the notes together and model several examples. Ask students to predict answers before checking them on their calculators.

• 25–35 minutes — Guided Absolute-Value Practice: Work through several absolute-value examples with students. Students enter examples into their calculators and record answers in their notes. Use questioning such as: 'What is the distance from zero?' and 'What happens when the number is negative?' Check for understanding before moving on.

• 35–45 minutes — Order of Operations Introduction: Transition to order of operations. Briefly review PEMDAS and explain why operations must be completed in the correct order. Model how to identify the first operation to complete and how to work through an

expression step by step.

• 45–55 minutes — Guided Order-of-Operations Examples: Complete several examples together. Students solve along with the teacher using their notes and calculators to check computations. Pause after each major step to ask students what should be completed next and why.

• 55–65 minutes — Independent Guided Examples: Students complete the remaining examples independently on their calculators. Teacher circulates, checks work, asks individual students to explain their steps, and reteaches or models a problem when needed. Review answers with the class before independent worksheets.

• 65–80 minutes — Independent Practice: Students complete two practice activities: an absolute-value maze and an order-of-operations color-by-number worksheet. Students work independently while the teacher circulates, provides feedback, and checks for understanding. Students may use calculators and their guided notes.

• 80–90 minutes — Closure/Review: Review selected problems from the worksheets and ask students to identify one strategy they used for absolute value and one strategy they used for order of operations. If time permits, finish with a short Blooket review using questions on absolute value and order of operations. If the Blooket cannot be completed, use a brief verbal/exit-ticket review instead.

Closure: What will you do to bring closure to the lesson? How will you summarize this lesson and preview the lesson that will follow?

Bring the lesson together by revisiting the essential question. Ask students: ‘How do you feel about what we learned today?’, 'What does absolute value tell us?' and 'Why do we need to follow the order of operations?'

Reflection Guidelines

The student will write lesson plans with an eye to the Danielson Framework and the Four Domains of Professional Practice. Use this as a resource when writing lesson plans.

Domain I Planning and Preparation

1. How do you know you were knowledgeable of your content?
2. What is the evidence that your objective was clear in the lesson plan?
3. How does the lesson address individual student learning needs?
4. What variety of strategies and methodologies did you consider when planning?
5. What is the evidence that lesson components were logically sequenced?
6. How did your lesson plan include checks for student understanding?
7. What PA Standards did this lesson address?

Domain II Classroom Environment

1. How did students know the behavioral and academic expectations for the lesson?
2. What effective routines and procedures did you implement for lesson efficiency?
3. How did you insure the classroom was physically accessible for all learners?
4. What evidence indicates that respect was shown between and among students and teacher?
5. What evidence indicates that students placed a high value on quality participation, process, and product?
6. How do you assess if the lesson was attainable for all students but also had appropriate rigor?

Domain III Instructional Delivery

1. How did you align your lesson to PA Standards?
2. How did you insure a clearly stated objective, understood by learners?
3. How did you provide clear explanations as to how to proceed through the lesson?
4. What questioning and discussion techniques insured all-learner participation?
5. What strategies did you employ to engage all students?
6. Describe how you made the content meaningful due to sequencing and pacing.
7. Discuss how you checked for understanding, provided feedback, retaught.
8. What is the evidence that you were you flexible and responsive to the needs of the learners?
9. Describe how you made meaning of the lesson with some form of closure activity.

Overall Assessment

Briefly state any changes you will make the next time you teach this lesson or activity, to improve any of the Domains of Professional Practice.